Replication of Temporal selection is suppressed, delayed, and diffused during the attentional blink by E Vul, M Nieuwenstein & N Kanwisher (2008, *Psychological Science*)

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Introduction

Vul et al., (2008) investigated how temporal selection of visual targets works, and along what dimensions it varies from one instance to the next using an attentional blink paradigm. In the attentional blink paradigm, participants fail to report the second of a pair of targets (T1 and T2) when they are presented at stimulus onset asynchronies (SOAs) of roughly 200 to 500 ms. Vul and colleagues directly tested the properties of temporal selection during the blink by analyzing distracter intrusions at a fast rate of item presentation. Their analysis of data from 12 participants showed that attentional selection is (a) suppressed (efficacy), (b) delayed (latency), and (c) diffused (precision) in time during the attentional blink as demonstrated by a significant three way interaction between SOA, measure (efficacy, latency, precision), and target with efficacy, latency and precision being dissociated by their time course such that each effect returns to baseline at different SOA.

An exact replication of Vul et al. (2008) will be conducted as part of the Reproducibility Project. 12 participants (Note 5 required to achieve 95% power based on original findings) will be tested using the same hardware and similar software specifications used in the original study. Original code used to run the experiment and analyze results will be used. The key finding for replication is the reported three way interaction of SOA, measure, and target.

Methods

Power Analysis

From data acquired from the authors and analyzed: Three way interaction (SOA, measure, target), partial eta^2 = .308, observed power = 1.000, R^2 = .765 (.744 adj), Cohen's d = 1.334. Power analysis to achieve 80%, 90%, 95% yields N = 4, 4, 5 respectively.

Planned Sample

Planned sample size for this experiment is 5 participants to achieve 95% power. Original demographics: “Twelve participants (7 female, 5 male) between the ages of 18 and 35 were recruited from the Massachusetts Institute of Technology subject pool and were paid for participation.” Note, however, that in consultation with the original authors it was recommended to run the same number of participants (12) as the task is quite difficult. Participants who cannot

perform the task during practice trials will be eliminated to avoid issues related to task difficulty. Participants recruited from The University of Western Ontario.

Materials

From the original text: *"On each trial, subjects saw an RSVP stream composed of one instance of each of the 26 English letters in a random order. Each letter was presented for 25 ms and was followed by a 58.3 ms blank, resulting in an RSVP rate of 12 items/s. The letters were white, uppercase, and in 48-point Courier font. They were presented on a black background. With our resolution (1024 x 768), monitor (Viewsonic G90f), and viewing distance (roughly 50 cm), the letters subtended roughly 2.51 of visual angle. On each trial, two cues in the RSVP stream indicated which of the letters were targets. The cues were white annuli with an inner diameter of 2.81 and an outer diameter of 3.21; thus, they appeared as rings around the cued letters. When a cue appeared, it was shown for 25 ms, concurrently with the target letter (see Fig. 2). Onset of the first cue (T1 onset) was randomly counterbalanced to occur equally often in Positions 6 through 10 in the RSVP stream. The delay between the first and second cues (T1-T2 SOA) was manipulated over 10 levels, from 1 to 10 items (83–833 ms). The experiment was programmed in Psychophysics Toolbox (Brainard, 1997; Matlab 7, Windows XP)."*

Note: This will be followed precisely. Original Matlab code supplied by the original experimenters.

Exceptions: A Viewsoic G90f will be used for the replication with Driver Version 1.4.0.28. A PC computer with Windows XP Version 2002, Service Pack 3, Matlab 7.2 and Psychophysics Toolbox Version 2.54 were used. The original description of the experiment requires a frame rate of 150Hz with a resolution of 1024 x 768. Multiple drivers were used in an attempt to get the monitor to present stimuli at 150Hz with 1024 x 768 resolution. This was not possible. After consulting with the original authors the protocol instructions listed setting the resolution of the screen to 640 x 480. This was also not possible. It was therefore decided in consultation with the original authors to run the experiment at 150Hz with a resolution of 800 x 600. Note that while these changes would affect the viewing angle of the stimuli, the original authors and the OSF reviewer do not think that this should affect the results of the study.

Procedure

The procedure is identical as reported in the original text: *"Each participant began the experiment with 2 practice trials that had a randomly selected T1-T2 SOA; the results of these trials were discarded. Following the practice trials, participants completed four blocks of 100 trials each. Each block contained 2 instances of each of the 50 possible combinations of T1 onset (five levels: Positions 6–10) and T1-T2 SOA (10 levels: SOAs of 1–10 items). For instance, in a trial with T1 onset at Position 6 and a T1-T2 SOA of 4 items, the two cues would appear during the 6th and 10th items in the sequence. These conditions appeared in a random order within each block. At the end of each trial, subjects were asked to indicate which two letters they thought had been cued by the annuli. Subjects reported the letters by pressing the corresponding keys on the computer keyboard. Duplicate letters were not accepted, and*

subjects were told to report the first letter first and the second letter second. After each trial, subjects were told how many points they had earned (1 point possible for each target). In addition to receiving the flat rate of $10 for participation, participants were offered bonus cash awards for performance: $0.01 for each letter reported correctly (on average, subjects reported about 200 letters correctly, receiving a $2 bonus). This bonus was provided as an incentive for subjects to try to report the cued letters correctly, given the particularly difficult conditions."

From the supplied protocol:

"AT5 protocol:
Experiment:
Booth C: C:\Progra~1\matlab\work\at5
Script: at5_test.m

1. *Set up room:*
    a. *Change screen resolution to 640x480.*
    b. *Refresh rate to 150 Hz.*
    c. *Matlab pwd to C:\Progra~1\matlab\work\at5*
2. *Explain demands of experiment:*
    a. *"Experiment will be about 1 hour long and involves you looking at a quickly flashing display and making responses by hitting a keyboard. You will be paid at least $10 for the hour, with the potential of making more. If you are uncomfortable, you can pause or stop your experiment at any time and still receive compensation. Please look over and sign the consent form."*
    b. *Get consent form signed, sign it as witness.*
    c. *Fill out row in subject log.*
3. *Bring into room, make sure subject turns off cell-phone/pager; explain where you will be (not in room) let subject get comfortable in front of screen.*
4. *run at5_test*
    a. *Session: 1 (unless practice, then 888)*
    b. *Subject: subjected > 1000, no repeats (if debugging, <1000)*
5. *Explain:*
    a. *<<press space>> "During this experiment you will be presented a rapid sequence of letters in the middle of the screen. Throughout the experiment, please look at the center of the screen, where you see the fixation point." <<press space>>*
    b. *"Each trial will begin with a brief fixation point. Then all 26 letters will be presented in a random order. They will be presented very quickly, with a brief blank between each one. Two of the blanks will contain a circular cue. Your job at the end of each trial is to report the two letters that were preceded by a cue. This task is designed to be extremely difficult, please do not be discouraged and try your best. Here is a sample trial:" <<press space>>*
    c. *(The prompt to enter letters appears after sequence) "Each trial ends by asking you to enter the two letters you saw. Press the appropriate letters on the keyboard. If you don't know, or know only one, you must guess anyway. You*

may sometimes need to hit the key twice for the response to register – make sure the response appears on screen. The order in which you enter letters doesn't matter. Please try to enter the letters from this trial now."

d. *(score comes up) "After each trial you will be told how many out of 2 possible letters you got right. The computer will also keep a running score for each block." <<press space>>*
e. *"Do you have any questions?"*
f. *"Now we will see an even more difficult trial." <<press space>>*
g. *(end of trial) "As you can see, this task can be extremely difficult and very frustrating. To keep you motivated in such strenuous conditions, we will pay you for how well you do. At the end of the experiment, you will be paid an additional penny for every point you get, totaled across all blocks. This means you could get as many as $8 dollars extra if you get them all right. You won't get that many right, since it is so difficult, but you will get some additional money, and the more you try, the better you will do, and the more money you will make. Do you have any questions?"*

"Each block will begin with a statement like the one you see now. When you are ready to begin the block, press the spacebar, and good luck.""

Note: This was followed precisely.

Exception: The supplied code is version AT6 while the supplied protocol sheet lists AT5. As noted above the resolution will be set to 800 x 600

Analysis Plan

From the original text: "*Our analyses are all based on the serial position of the reported letters at different T1-T2 SOAs. We computed the distribution of reports for T1 from the first responses and the distribution of reports for T2 from the letters reported second... We estimated efficacy of selection as the average probability of reporting an item from a seven-item window around the target (spanning three items before to three items after the target. This efficacy measure (see the appendix) is independent of delay and diffusion, as it does not depend on which item is reported (the target or earlier or later items within the window)... The second relevant metric of temporal selection is its latency, which we estimated as the center of mass of reports in the window around a given target. This measure (originally employed by Chun, 1997a) corresponds to the average reported serial position, relative to the target (see the appendix), and is independent of both the efficacy and the precision of selection... A third metric of selection is the precision of selection around the center of mass. This can be measured by the variance of the center of mass of reported items (see the appendix)... To assess whether the three measures of selection follow different time courses, we first conducted an analysis of variance with the factors of SOA (1–10 items), measure (efficacy, latency, or precision), and target (1 or 2).*

Appendix: We calculated A, the average probability of report (efficacy of selection: our measure of suppression), as follows:

$$A = \frac{\sum_{i=ks}^{ke} P_i}{n},$$

where n is the total number of items in the selection window (in our case, 7), Pi is the probability (empirical frequency) of reporting an item from serial position i (relative to the target position, 0), and ks and ke are the lower and upper bounds, respectively, of the window used to compute the measure, expressed in serial position of the item relative to the cue position. We used ks 5 _3 and ke 5 3 (0 is the correct target). C, the center of mass (latency of selection: our measure of delay), was calculated as

$$C = \frac{\sum_{i=ks}^{ke} P_i * i}{A * n}.$$

Finally, V, the variance of the center of mass (precision of selection: our measure of diffusion), was calculated as"

$$V = \frac{\sum_{i=ks}^{ke} P_i * (i - C)^2}{A * n}.$$

Note: This will be followed precisely. Original Matlab analysis code supplied by the original experimenters will be used to first confirm reported results of the original study using the original data set. This code will then used to analyze the replication results.

Differences from Original Study

As noted above, the original description of the experiment requires a frame rate of 150Hz with a resolution of 1024 x 768. This was also not possible. It was therefore decided in consultation with the original authors to run the experiment at 150Hz with a resolution of 800 x 600. In principle this change in resolution and corresponding change in the visual angle of the stimulus should not make a difference based on claims in the original article or subsequent published research on the conditions for obtaining the effect. Both the original authors and the OSF reviewer agree with this assessment.

(Post Data Collection) Methods Addendum

Actual Sample

Twelve participants (7 female, 5 male) between the ages of 18 and 27 were recruited from Western University and were paid for participation. All participants provided informed written consent to participate in the experiment which was approved by The University of Western Ontario's research ethics board which complies with the declaration of Helsinki.

Differences from pre-data collection methods plan

none

Results

Data preparation

Data preparation following the analysis plan.

Confirmatory analysis

Analyses were conducted as specified in the analysis plan. As the original data and analysis source code was supplied by the original authors we can directly compare the original results from Vul et al (2008) shown in Table 1, as compared to the present replication as shown in Table 2. Figures 1 and 2 also for a direct comparison of the two studies.

Analysis of Variance

Source	Sum Sq.	d.f.	Mean Sq.	F	Prob>F
X1	169.392	2	84.6961	602.76	1.21636e-149
X2	9.433	1	9.4334	67.14	1.31678e-15
X3	9.444	9	1.0493	7.47	1.94008e-10
X1*X2	6.73	2	3.3649	23.95	9.11802e-11
X1*X3	57.344	18	3.1858	22.67	1.86475e-57
X2*X3	8.016	9	0.8906	6.34	1.21658e-08
X1*X2*X3	41.255	18	2.292	16.31	6.00903e-42
Error	92.739	660	0.1405		
Total	394.353	719			

Constrained (Type III) sums of squares.

Table 1: Original results as reported by Vul et al (2008) as generated from original analysis source code and data. Target for replication are significant main effects of efficacy (X1), latency (X2) and precision (X3), as well as their three-way interaction: $F(18, 660) = 16.31$, $p < .00001$.

Analysis of Variance

Source	Sum Sq.	d.f.	Mean Sq.	F	Prob>F
X1	334.485	2	167.243	587.97	2.3759e-147
X2	18.802	1	18.802	66.1	2.1207e-15
X3	11.26	9	1.251	4.4	1.33372e-05
X1*X2	25.243	2	12.621	44.37	8.2998e-19
X1*X3	69.801	18	3.878	13.63	5.96108e-35
X2*X3	10.34	9	1.149	4.04	4.72067e-05
X1*X2*X3	66.468	18	3.693	12.98	3.30113e-33
Error	187.73	660	0.284		
Total	724.129	719			

Constrained (Type III) sums of squares.

Table 2: Results of replication as generated using the original analysis source code supplied by Vul et al (2008). All main effects for efficacy (X1), latency (X2) and precision (X3) were found significant as well as their three-way interaction: $F(18, 660) = 12.98$, $p < .00001$.

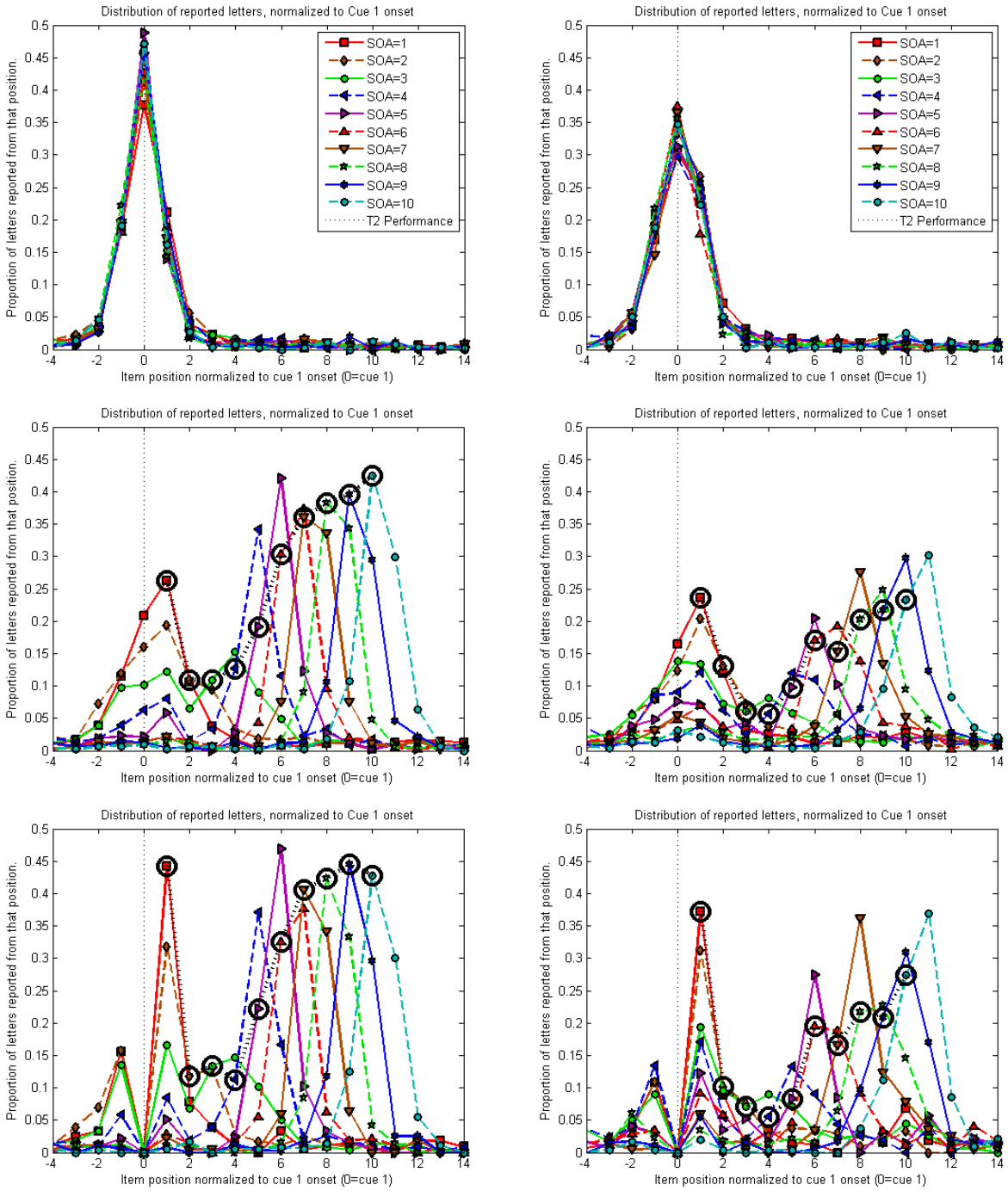


Figure 1: Original results from Vul et al (2008; Figure 3) as generated from original analysis source code and data (left), and replication results (right). From Vul et al: *Probability of reporting a letter from a particular position in the rapid serial visual presentation stream. Results are shown separately for (a) the first target (T1), (b) the second target (T2), and (c) T2 given that T1 was reported correctly. The thick black lines in (b) and (c) indicate the probability of accurate T2 report as a function of stimulus onset asynchrony (SOA). Item position is normalized to T1 (i.e., T1 is always at x 5 0).*

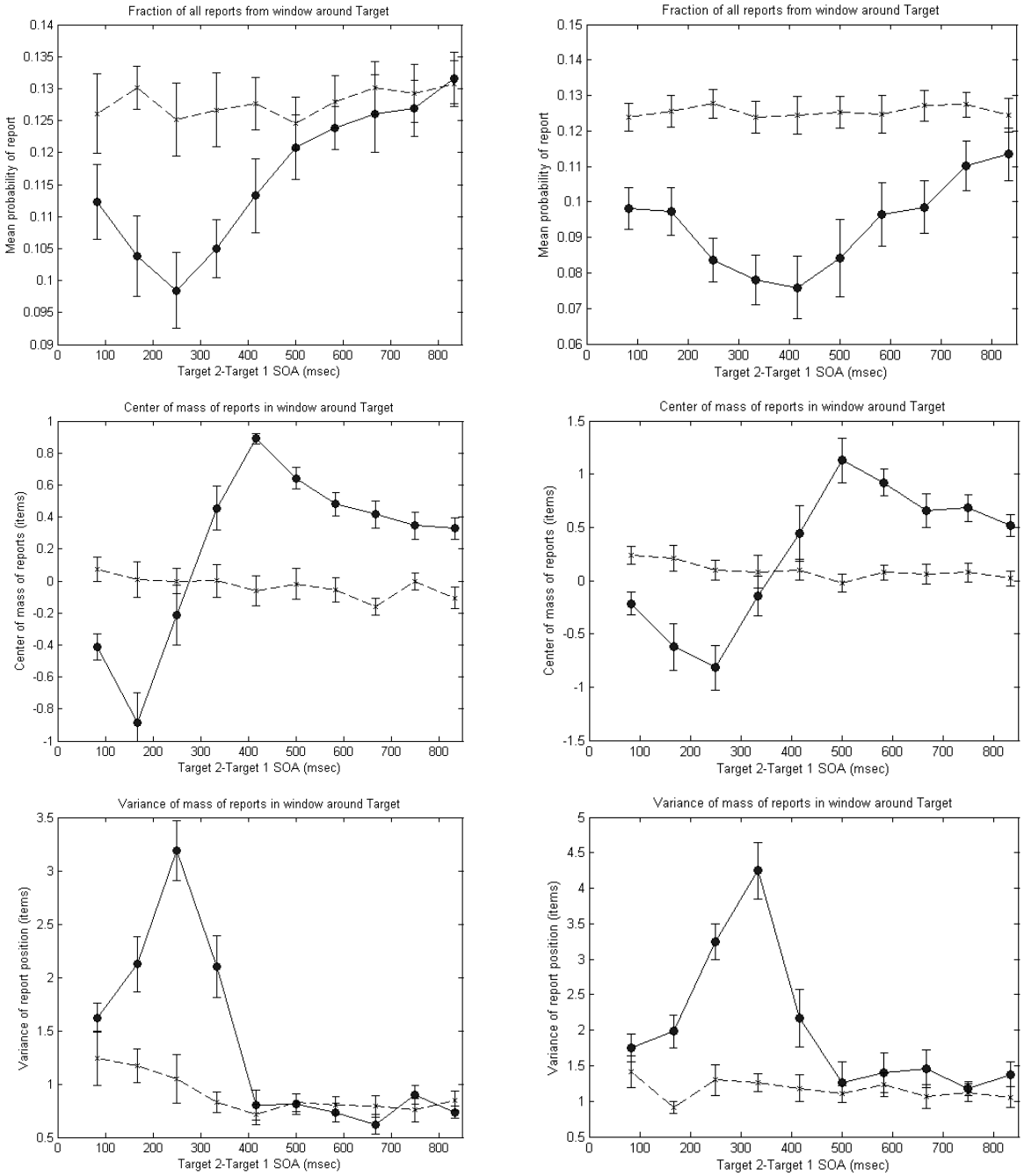


Figure 2: Original results from Vul et al (2008; Figure 4) as generated from original analysis source code and data (left), and replication results (right). Note the change in scale on the y axes between the original and replicated results. From Vul et al: *Results for the three selection-window measures: (a) average probability of reporting an item within the seven-item selection window around the target (suppression), (b) center of mass of reports in the window around the target (delay), and (c) variance of the center of mass in the window around the target (diffusion). Each measure is shown as a function of stimulus onset asynchrony (SOA) between the two targets. All error bars correspond to 1 SEM.*

Discussion

Summary of Replication Attempt

The analysis confirms the original effects observed by Vul and colleagues (2008), namely that selection is (a) suppressed (efficacy), (b) delayed (latency), and (c) diffused (precision) in time during the attentional blink as demonstrated significant main effects of each of these factors and by a significant three way interaction between SOA, measure (efficacy, latency, precision). The other main finding by Vul et al (2008) was that efficacy, latency and precision can be dissociated by their time course such that each effect returns to baseline at different SOA. These effects were also found as illustrated by comparing the figures from the original study with the replication results in Figures 1 and 2. While there are subtle differences in the specific results for the effects of suppression, delay and diffusion, the overall pattern of results is essentially the same between the original and replicated results. In sum the results from the original study by Vul et al (2008) were successfully replicated.